Toward a Theory of Computation for Learned Transformer Machines: Tasks, Resources, Learnability, Composition, and Closure

Paper 0.1 — Position and Formal Framework

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Abstract

Classical expressivity asks what an idealized Transformer can compute; benchmark studies ask whether a trained model succeeds. Between them lies the theory of a *learned* Transformer machine. We formalize task structure, information sufficiency, machine resources, acquisition probability, one-pass and autoregressive composition, and resource-scheduled closure. The operational object is compound, $\mathcal{M} = (F_\theta, C, S, I, \mathbb{T}, \mathcal{B})$. Five executable fixed-weight witnesses calibrate representability under explicit finite encodings. Their contrast with Paper 0.85—where trained one-token and autoregressive machines share frontier $K = 3$ and fail at unseen $K = 4$—makes the central distinction concrete: representability is not reliable acquisition. Empirical papers appear only as late, nonessential anchors.

1 Introduction

A fixed architecture may represent a computation that training rarely discovers; a trained model may fit bounded instances without a reusable rule; a reusable local rule may still fail under composition or termination. Autoregression changes the effective machine by applying the same transition repeatedly. We therefore separate

$$\boxed{\text{representability} \neq \text{learnability} \neq \text{reliable acquisition} \neq \text{systematic executability} \neq \text{closure.}}$$

The evidential layers are distinct: propositions follow from premises; executable hand-weighted constructions establish finite-domain representability; trained-model experiments measure acquisition; conjectures remain prospective.

2 Task families and concrete examples

A task family is $\mathcal{T} = (\mathcal{X}, \mathcal{Y}, \mathcal{D}, \mathcal{R})$, where $\mathcal{D}$ is a generator/distribution and $\mathcal{R}$ the declared prior. Its partially ordered complexity vector is

$$\mathbf{c} = (s_{\text{atom}}, k_{\text{comp}}, b_{\text{branch}}, v_{\text{state}}, n_{\text{nuisance}}, m_{\text{context}}, \ldots).$$

The coordinates measure atomic syntax, composition/proof depth, branching, state, nuisance, and context/memory demand. Raw length need not increase composition depth.

2.1 From concrete tasks to the abstract task vector

T1: atomic mapping. `A->B, B->C, QUERY B` targets `C`: very low syntax, $k = 1$, unit branching, negligible state/nuisance.

T2: contextual lookup. `A->X, B->Y, C->Z, QUERY B` targets `Y`: low syntax and $k = 1$, but contextual cross-token transport. Distractors can increase length without depth.

T3: grandparent. `PARENT C B, PARENT B A, QUERY GRANDPARENT C` targets `A`: parent lookup composed twice.

T4: root. For `C->B->A->R`, a one-token answer needs bounded internal composition; multi-token `C B A R` needs recurrence and termination for scalable execution.

T5: ancestor-(k). `ANCESTOR C 2` needs node-and-counter state, decrement, and termination.

T6: proof chain. From `A`, `A->B->C->D`, decide `D?`; proof depth, distractors, and branching vary independently.

T7: unification. `P(f(x),y)` with `P(f(a),b)` yields $\{x = a, y = b\}$: high syntax, binding state, and functor depth.

T8: branching search. `A->B,A->C,B->D,C->E,...` separates search width and frontier/visited state from proof depth.

Task	s_{atom}	k_{comp}	b_{branch}	v_{state}	Context	Recurrence
Successor	low	1	1	low	optional	no
Pair lookup	low	1	1	low	yes	no
Grandparent	low	2	1	low	yes	no
Root	low	variable	1	current node	yes	useful
Ancestor-(k)	low	variable	1	node+counter	yes	yes
Proof chain	medium	variable	variable	proof state	yes	useful
Unification	high	bounded	1	bindings	yes	maybe
Branching search	medium	variable	high	frontier+visited	yes	yes

Table 1: Concrete tasks mapped to the abstract vector; entries are task requirements or useful protocols, not architectural lower bounds.

3 Information sufficiency

For query X, context D, target Y, and prior $\mathcal{R}$, validate

$$I(Y; D \mid X) = H(Y \mid X) - H(Y \mid X, D), \qquad H(Y \mid X, D, \mathcal{R}) = 0$$

for deterministic identifiable tasks. Record

$$\mathcal{I} = (H(Y \mid X), H(Y \mid X, D), I(Y; D \mid X), \mathcal{R}, A_{\text{Bayes}}, |\mathcal{H}_{X,D,\mathcal{R}}|, m_{\text{id}}).$$

These mean, respectively: query-only uncertainty; remaining uncertainty after context; context's added information; declared hypothesis family; best possible expected accuracy; number of consistent hypotheses; and minimum context for identification.

Impossible random mapping. `A->7, B->2, QUERY C` is underdetermined if C's output is independent: $H(Y \mid X, D) > 0$.

Permutation completion. `A->3, B->1, C->4, QUERY D` forces `D->2` only under the prior that outputs permute $\{1, 2, 3, 4\}$. This is constraint completion, not general induction.

Affine induction. Under $f(x) = ax + b \pmod 7$, $f(0) = 3, f(1) = 5$ give $b = 3, a = 2$. The prior changes identifiability.

Standard Bayes-risk fact. Positive conditional entropy on positive measure implies $A_{\text{Bayes}} = \mathbb{E} \max_y P(Y = y \mid X, D, \mathcal{R}) < 1$. This routine fact enforces dataset discipline; it is not a novelty claim.

The zero-conditional-entropy condition applies to deterministically identifiable tasks, which Papers 0.5–0.9 primarily study; it is not a universal requirement for useful prediction. In a probabilistically predictive task, $H(Y \mid X, D, \mathcal{R}) > 0$: context can reduce uncertainty, $I(Y; D \mid X) > 0$, without eliminating it. Evaluation must then declare a null predictor, measure improvement over it, and compare with the Bayes-optimal distribution. This is a secondary evaluation regime, not a seventh computational axis; Appendix B gives the details.

4 Machine, controller, and resource formalism

C_0 invokes F_θ once; C_1 invokes it autoregressively; C_2 gives state typed fields; C_3 is an external action/tool loop.

Definition 1 (Compound learned machine)**.**

$$\mathcal{M} = (F_\theta, C, S, I, \mathbb{T}, \mathcal{B}), qquad P_{\text{acquire}} = P_\Pi[A(\tau, \mathbf{c}; \mathcal{M}, \Pi) \geq \gamma].$$

F_θ is the learned transition map, C its scheduling protocol, S writable state, $I : q \mapsto D$ information, $\mathbb{T}$ partial external operations, and $\mathcal{B}$ context/call/token/time/memory budget.

Report $\Delta\mathcal{M} = (\Delta C, \Delta S, \Delta I, \Delta\mathbb{T}, \Delta\mathcal{B}, \Delta\Pi)$, so easier prompts, stronger tools, more time, and different training are not all "augmentation."

5 Six computational axes

1. **Atomic acquisition:** which local operators pass across seeds?
2. **One-pass composition:** $K_{\text{internal}} = \max\{K : A(f^K) \geq \gamma\}$, controlling dependency radius, depth, nuisance, and length.
3. **Autoregressive composition:** $s_{t+1} = F_\theta(X, s_{\leq t})$; report transitions, stopping, sequence success, and train-short/test-long behavior.
4. **Closure signature:** separate a contiguous tested frontier from a uniform resource-scheduled procedure.
5. **In-context information:** separate copying, completion, task selection, parameter estimation, and procedure induction.
6. **Tool/harness augmentation:** separate computational, acquisition, reliability, and efficiency gains.

Much classical circuit/logical work studies a fixed one-pass Transformer under precision assumptions. It cannot be transferred automatically to C_1, where emitted tokens create serial applications of the transition map.

6 Constructive witnesses and the learnability gap

Representability asks whether $\exists W^*$ such that $\mathcal{M}(W^*)$ implements τ; acquisition asks for $P_\Pi[\mathcal{M}(W_\Pi)$ passes $\tau]$. The hand-weighted witnesses instantiate the first statement without training.

Witness	Protocol	((L,H,d))	Legal result	Main control	Scope
Successor	C_0	$(1, 0, 4)$	3/3, margin 1	SA 0/3	local weight map
Pair lookup	C_0	$(1, 1, 9)$	18/18	FF 6/18 ties	atomic pairs
Grandparent	C_0	$(2, 1, 20)$	2/2	one-layer SA 0/2	fixed two-hop chain
Root	C_1	$(1, 1, 15)$	24/24	FF 6/24	depth 0–3
Unary chain	C_1	$(1, 1, 15)$	96/96	FF 24/96	length 1–4

Table 2: Fixed-weight calibration. Exact means exhaustive argmax correctness, not one-hot attention; no witness is claimed minimal.

Paper 0.85 supplies the learned contrast. On its corrected dataset, one-token O0, free-trace O1, and supervised structured O3 have contiguous frontier three; O2 has none. Every condition has zero three-seed acquisition probability at $K = 4$. Thus $K_{\text{internal}} = K_{\text{AR}} = 3$ in that phase despite a handcrafted recurrent transition under a different atomic encoding: representability does not imply acquisition.

7 Composition and resource-scheduled closure

For legal set R_K, per-K solvability $\forall K \exists(\mathcal{M}_K, \mathcal{B}_K)$ differs from

$$\exists\mathcal{M}\exists\mathcal{B}(K)\forall K < \infty : \ \mathcal{M} \text{ solves all } x \in R_K \text{ within } \mathcal{B}(K).$$

Empirically, K_{prefix} is the largest k for which every tested $1 \leq j \leq k$ passes. Report

$$\Sigma = (K_{\text{train}}, K_{\text{test}}, K_{\text{prefix}}, \rho_{\text{step}}, \rho_{\text{stop}}, g, f_{\text{fail}}).$$

Proposition 1 (Finite observations do not establish closure)**.** *For any finite evaluated input–output set consistent with a resource-scheduled procedure, a finite lookup machine agrees on all observations and diverges on an unobserved legal input.*

Proposition 2 (Budgeted local composition)**.** *If D is closed under deterministic $T : D \to D$, and C exactly preserves state, invokes/integrates T, and applies a total correct stop rule, then $T^k(x)$ is representable whenever all states and trace fit $\mathcal{B}$.*

Independent reliability $r_{\text{stop}} \prod_i r_i$ is only a baseline; acquisition and reliability are separate.

8 Information interfaces and tool algebra

Retrieval changes evidence: $q \mapsto I(q) = D \mapsto F_\theta(q, D)$. Tools are semantic: T_0 lookup, T_1 local transition, T_2 transform, T_3 search/closure, T_4 programmable executor.

	C_0	C_1	C_2	C_3
No tool	one-shot	generation	typed scratchpad	environment loop
T_0	evidence	retrieve/generate	evidence state	iterative retrieval
T_1	one call	serialized calls	current/counter	parent iteration
T_2	delegated transform	calculator/PAL	expression/result	CAS/SQL
T_3	delegated answer	closure call	query/result	search
T_4	run code	generate/execute	program state	code agent

Table 3: Controller $\times$ tool configurations, not an intelligence ladder.

For information I_s, acquisition Q, and execution E, $P(I_s \cap Q \cap E) = P(I_s)P(Q \mid I_s)P(E \mid Q, I_s)$. This equals endpoint success only when shortcut branches are excluded. The heterogeneous-network extension is developed separately in Paper 0.2.

9 Classical theory and related work

Topic/work	Pass	Assumptions	Model	Evidence	Boundary here
Hahn; Hao	fixed	attention restrictions	encoder	theory	acquisition
Merrill–Sabharwal; Chiang	fixed	log precision	encoder	theory	learned frontier
Strobl survey	fixed	variant-dependent	both	survey	schema
Merrill–Sabharwal CoT; Li	AR	serial tokens	decoder	theory	acquisition
Joshi; Amiri Bavandpour	AR	iterated/hard attention	decoder	theory	finite learned K
Elhage; Olsson; Meng	fixed activations	trained finite	decoder	mechanistic	causal vs geometric
Dziri; Huang	trained	distribution-specific	decoder	empirical/theory	task controls

Table 4: Topic-organized comparison: assumptions do not transfer across pass, precision, or encoder/decoder setting.

Fixed one-pass. Results depend on attention, precision, positions, and uniformity [1, 2, 3, 4, 5, 6, 7]; universality is representability, not acquisition [8, 9].

Autoregressive/CoT. Serial tokens change expressivity under explicit assumptions [10, 11, 12, 13].

Mechanistic circuits. Residual decomposition, induction heads, and causal tracing distinguish description from necessity [14, 15, 16].

Composition limits. Endpoint success can conceal brittle, distribution-specific computation [17, 18].

10 Late empirical anchors

Paper	Axis	Status/illustration
0.5	internal composition	evidence; non-monotone depth, distributed heads
0.6	semantic frontier	evidence; budget–allocation interaction
0.7	formal inference	infrastructure; learned frontier pending
0.8	information/ICL	evidence; sufficiency vs acquisition
0.85	AR recurrence	Stage 1; no $K = 4$ acquisition or AR gain
0.9	tools/harness	pending

Table 5: The framework does not depend on these bounded series results.

11 Conjectures and status

	Conjecture	Status
C1	local acquisition need not imply composition	consistent with 0.6/0.85
C2	trained models often have bounded one-pass frontiers	consistent with 0.5/0.6
C3	multi-token state expands the measured frontier	not supported in 0.85 Stage 1; construction is representability only
C4	competent recurrence shifts failure toward control	pending
C5	context separates knowledge from procedure	consistent as design principle/0.8 taxonomy
C6	harnesses may improve reliability before expressivity	pending 0.9

Table 6: Falsifiable conjectures with bounded status language.

12 Discussion and conclusion

Validate identifiability; establish local competence; measure contiguous composition and recurrence frontiers; only then interpret mechanism. Fixed weights calibrate representability, while learned experiments measure acquisition. Resource-scheduled closure is a quantified property of one reusable procedure, not a long successful trace.

A Fixed-weight micro-Transformer constructions

All implementations use row vectors XW, not WX; attention is finite $\mathrm{softmax}(QK^\top/\sqrt{d_h})V$. Complete artifacts are under `results/manual_witnesses/`.

A.1 FF-only successor

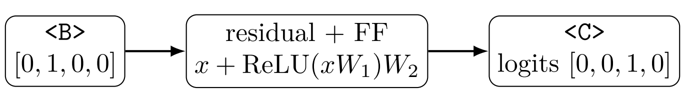

Figure 1: Successor, margin 1.

$$M = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad W_1 = I_4, \quad W_2 = M - I_4 = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

For legal one-hots, $x + \mathrm{ReLU}(xW_1)W_2 = xM$. Result/control: 3/3; SA 0/3; SA+FF 3/3. Biases, positions, and attention are zero.

A.2 SA-only pair lookup

Architecture $L = 1, H = 1, d = 9, d_h = 3$, subspaces [key(3)|query(3)|value/output(3)].

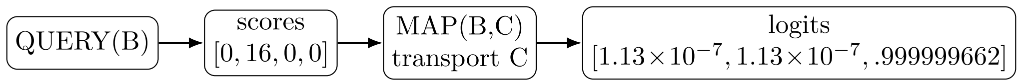

Figure 2: Finite key match and value transport.

Actual sparse blocks: $W_Q[4{:}6, :] = 27.7128129211 I_3 = 16\sqrt{3} I_3$, $W_K[1{:}3, :] = I_3$, $W_V[7{:}9, :] = I_3$, $W_O[:, 7{:}9] = I_3$, others zero. Probabilities are $(1.125351{\times}10^{-7}, .999999662395, 1.125351{\times}10^{-7}, 1.125351{\times}10^{-7})$; margin .999999549859. Result/control: 18/18; FF 6/18 ties; SA+FF 18/18.

A.3 Two-layer SA grandparent

Architecture $L = 2, H = 1, d = 20, d_h = 4$; subspaces are edge key, initial query, parent value, hop state, output.

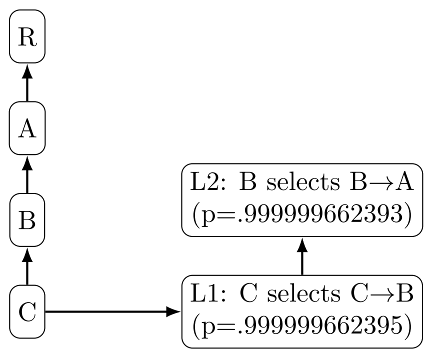

Figure 3: First hop is written to state; layer 2 reads it.

Layer 1 maps query $\to Q$, edge key $\to K$, and parent value $\to V \to$ hop state; layer 2 maps hop state $\to Q$, the same edge blocks to K, V, and writes output. Canonical selected probabilities are .999999662395 and .999999662393; final logits are $(1.125359 \times 10^{-7}, .999999662393, 1.125357 \times 10^{-7}, 0)$, margin .999999549857. Result/control: 2/2; FF and one-layer SA 0/2; SA+FF 2/2.

A.4 Autoregressive root

Architecture $L = 1, H = 1, d = 15, d_h = 5$, reused by an external C_1 model loop.

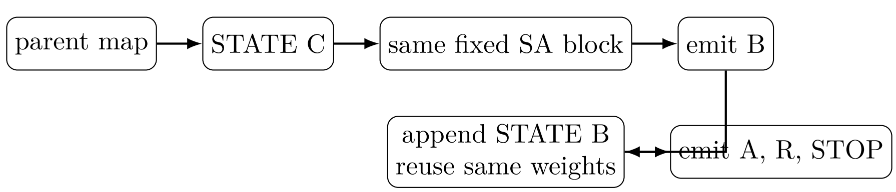

Figure 4: The driver appends argmax tokens but performs no semantic transition.

Step	Current	Target/prediction	Attention (p)	signed margin
1	C	B/B	.999999549860	.999999437324
2	B	A/A	.999999437324	.999999324789
3	A	R/R	.999999324789	.999999212254
4	R	STOP/STOP	.999999212254	.999999099719

Table 7: Tracked canonical trace.

Canonical, reversed, and fixed-shuffle orders each pass all 24 root cases; minimum margin .999999099719. FF passes 6/24 ties; SA+FF 24/24. Cycles, missing/duplicate/conflicting records, non-atomic encodings, larger vocabularies, and out-of-budget lengths are excluded.

A.5 Artifact and formal details

Complete Q/K, raw $QK^\top$, score, probability, V, residual, FF, and logit traces, signed and winner margins, matrices, legal rows, and controls are documented in `manual_transformer_witnesses.md` and generated by `src/cl/manual_transformers/`.

B Probabilistic tasks and better-than-null competence

B.1 Deterministic and probabilistic regimes

Let $P^*(Y \mid X, D, \mathcal{R})$ be the true conditional law under the declared generator and prior, $q_\theta(Y \mid X, D)$ the machine distribution, and q_0 an explicit admissible null. A null may be uniform, the class-frequency or majority prior, a query-only predictor $q_0(Y \mid X)$, a context-shuffled predictor, or a justified task-specific baseline. "Chance" is not a specification. Query-only versus context-aware performance tests whether the machine uses predictive information in D; mutual information in the generator does not by itself establish such utilization.

Define the null and machine accuracies under their declared decision or sampling rules and

$$A_{\text{Bayes}} = \mathbb{E}_{X,D} \max_y P^*(Y = y \mid X, D, \mathcal{R}).$$

A useful stochastic predictor may have $A_0 < A_\theta < A_{\text{Bayes}} < 1$. This is better-than-null predictive competence, not deterministic task solution, rule identification, compositionality, recurrence, closure, or a causal mechanism.

Regime	Entropy condition	Primary criterion	Appropriate claim
Identifiable	$H(Y \mid X, D, \mathcal{R}) = 0$	exact accuracy/coverage; ceiling 1	task competence
Probabilistic	$H(Y \mid X, D, \mathcal{R}) > 0$	log loss and accuracy versus null; Bayes ceiling may be < 1	predictive competence
Repeated/ensemble	same underlying task	$A(n, G)$, correlation, cost	compound predictive competence
Oracle best-of-n	either	candidate coverage C_n	coverage only

Table 8: Task/evaluation regimes orthogonal to the six computational axes.

B.2 Proper scoring, calibration, and uncertainty

Accuracy discards distribution quality. For a known generator,

$$\mathrm{CE}(q) = -\mathbb{E} \log q(Y \mid X, D), \qquad \Delta_{\mathrm{CE}} = \mathrm{CE}(q_0) - \mathrm{CE}(q_\theta).$$

$\Delta_{\mathrm{CE}} > 0$ means better probabilistic prediction than the null (in nats when log is natural). One may also report $D_{\mathrm{KL}}(P^* \| q_\theta)$ only when P^* is known or has a justified estimator. On finite outcomes, NLL, Brier score, and reliability curves complement accuracy; ECE depends on binning and is not required for the deterministic proof/tree experiments.

For illustration, let $P^* = (.50, .25, .15, .10)$, $q_0 = (.25, .25, .25, .25)$, and $q_\theta = (.45, .25, .20, .10)$. The machine assigns outcome 1 probability .45 rather than .25. Independent target and prediction sampling gives match accuracy .3275 versus .25; deterministic argmax accuracy is $.50 = A_{\text{Bayes}}$. In nats, $\mathrm{CE}(q_0) = 1.38629$, $\mathrm{CE}(q_\theta) = 1.21750$, and $\Delta_{\mathrm{CE}} = .16879$. These illustrative numbers expose why accuracy alone misses calibration; they are not experimental results.

Finite evaluations report N and uncertainty, preferably a paired comparison on shared instances; acquisition claims additionally require training seeds. Exhaustive tiny legal domains should use exhaustive accuracy. A capability ladder is

$$\text{null-level} < \text{better-than-null} < \text{Bayes-optimal probabilistic},$$

followed by Bayes-optimal deterministic identification only when the task permits it.

B.3 Repetition, ensembles, and correlated errors

For a declared stochastic mechanism, draw $Y^{(1)}, \dots, Y^{(n)}$ and aggregate

$$\widehat{Y}_n = G_n(Y^{(1)}, \dots, Y^{(n)}), \qquad A(n, G) = P(\widehat{Y}_n = Y).$$

Report both $A(1)$ and $A(n, G)$. Majority/plurality, probability averaging, consensus thresholds, and verifier selection are system-level protocols. Distinguish same-model stochastic repetition, independent trained seeds, architecture/model ensembles, and context/data perturbations; their error correlations differ.

For binary independent runs with odd n and per-run correctness $p > .5$,

$$A_{\text{maj}}(n) = \sum_{j=(n+1)/2}^{n} \binom{n}{j} p^j (1-p)^{n-j} \longrightarrow 1.$$

This is intuition under iid assumptions, not a forecast for language-model samples; correctness-majority also coincides with label-majority only for binary outcomes or a single wrong label. Runs can share a wrong interpretation, shortcut, training bias, prompt-induced failure, or decoding mode. Useful diagnostics include pairwise answer agreement, correctness correlation, answer-histogram entropy, effective answer count, and the saturation curve $A(n)$. Repeated identical wrong answers are not stronger evidence. If $q_\theta = q_0$, there is no side-information verifier, and aggregation is label-symmetric, repetitions with no complementary information create no missing task information.

B.4 Candidate coverage and deployable selection

Oracle best-of-n measures only

$$C_n = P\Big(Y \in \{Y^{(1)}, \dots, Y^{(n)}\}\Big),$$

the probability that a correct candidate appears. Deployable accuracy requires an aggregator or verifier to select without target ground truth. High C_n can coexist with poor $A(n, G)$; report both and the selector's cost and information. Repetition plus aggregation, $\mathcal{M}^{(1)}, \dots, \mathcal{M}^{(n)} \to G$, is already a simple compound computation. Paper 0.2 generalizes this narrow bridge to heterogeneous nodes with different capabilities, costs, reliabilities, and interfaces.

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